AI BASED FINGERPRINT AND IRIS PRIVACY PROTECTION SYSTEM

A PROJECT REPORT

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in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)





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OCTOBER 2026

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BONAFIDE CERTIFICATE

Certified that this Project report titled "**AI BASED FINGERPRINT AND IRIS PRIVACY PROTECTION SYSTEM**" is the bonafide work of "**RITHUPRIYA.S (2116231901043), SHAMRUDHA VARSHINI.P (2116231901048)**" who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

The increasing use of social media has made sharing high-resolution photographs a common part of everyday digital communication. However, such images may unintentionally expose sensitive biometric information, particularly fingerprints and iris patterns. Since biometric characteristics are unique to individuals and cannot be easily changed once compromised, their accidental exposure can create significant privacy and security concerns. Existing biometric security systems mainly focus on authentication, recognition, or protection of stored biometric templates, while less attention is given to identifying biometric exposure in ordinary photographs before they are shared online.This project proposes an AI-Based Fingerprint and Iris Privacy Protection System, named BioShield, to assess potential biometric exposure in images before social-media sharing. The proposed system uses computer-vision techniques to identify potential fingerprint and iris regions and analyzes their visibility and image quality based on factors such as resolution, sharpness, contrast, and region size. A Vision Transformer (ViT) is proposed as the primary deep-learning model for analyzing the extracted biometric regions and assessing their potential privacy exposure. The system highlights potentially exposed biometric regions and provides users with an understandable explanation of the associated privacy risk before the image is shared. The proposed framework does not perform biometric authentication, identity recognition, or biometric matching. Instead, it focuses on preventive privacy assessment and provides the analysis required for subsequent selective protection of sensitive regions. The project aims to improve user awareness of biometric exposure and provide a practical cybersecurity approach for safer sharing of digital photographs.

ACKNOWLEDGEMENT

First, we thank the almighty God for the successful completion of the project. Our sincere thanks to our Chairman **Mr. S. Meganathan B.E., F.I.E.,** for his sincere endeavor in educating us in his premier institution. We would like to express our deep gratitude to our beloved Chairperson **Dr. Thangam Meganathan Ph.D.,** for her enthusiastic motivation which inspired us a lot in completing this project and Vice Chairman **Mr. Abhay Shankar Meganathan B.E., M.S.,** for providing us with the requisite infrastructure.

We also express our sincere gratitude to our college Principal **Dr. S.N. Murugesan M.E., Ph.D.,** and **Dr. P. Tamilselvi Ph.D., Professor and Head, Department of Computer Science and Engineering (Cyber Security)** and our project guide **Dr. Bhuvana S Ph.D., Assistant Professor (SG)** for her encouragement and guiding us throughout the project. We would like to thank our project coordinator **Mr. Benedict J.N., M.E., Assistant Professor (SG),** for his encouragement and guidance towards successful completion of this project and to our parents, friends, all faculty members and supporting staffs for their direct and indirect involvement in successful completion of the project for their encouragement and support.

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TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	iii
	LIST OF TABLES	vii
	LIST OF FIGURES	viii
1	**INTRODUCTION**	**1**
	1.1 BACKGROUND	1
	1.2 PROBLEM STATEMENT	1
	1.3 OBJECTIVES	2
	1.4 SCOPE	3
	1.5 SECURITY REQUIREMENTS/ SECURITY CONSIDERATIONS	3
	1.6 SDG ALIGNMENT	3
2	**LITERATURE REVIEW**	**5**
	2.1 EXISTING SYSTEM WITH SECURITY APPROACHES	5
	2.2 RESEARCH GAPS	5
	2.3 PROPOSED SYSTEM	6
	2.3.1 OVERVIEW OF THE PROPOSED SYSTEM	6
	2.3.2 OBJECTIVES OF THE PROPOSED SYSTEM	7
	2.3.3 ADVANTAGES OF PROPOSED SYSTEM	7
3	**SYSTEM DESIGN**	**8**
	3.1 SYSTEM ARCHITECTURE	8
	3.2 MODULE DESIGN	9
	3.3 DATA FLOW/UML DIAGRAMS	10
	3.4 THREAT MODEL AND ATTACK SURFACE	12
	3.5 SECURITY CONTROLS	13

4	**SYSTEM DESCRIPTION**	**14**
	4.1 DATA COLLECTION	14
	4.2 DATA PREPROCESSING	14
	4.2.1 ALGORITHMS / METHODOLOGY	14
	4.2.2 RESULTS AND DISCUSSION	16
5	**CONCLUSION**	**18**
	REFERENCES	**19**
	APPENDIX	**21**

LIST OF TABLES

TABLE NO.	TITLE	PAGE NO.
1.1	SDG ALIGNMENT OF THE PROPOSED PROJECT	3
2.1	POSITIONING RELATIVE TO EXISTING APPROACHES	6
3.1	HARDWARE AND SOFTWARE REQUIREMENTS	9
4.1	PRIVACY RISK CLASSIFICATION	16

LIST OF FIGURES

FIGURE	TITLE	PAGE NO.
3.1	SIMPLIFIED SYSTEM ARCHITECTURE	8
3.2	DETAILED COMPONENT-LEVEL ARCHITECTURE	9
3.3	USE CASE DIAGRAM	11
3.4	ACTIVITY DIAGRAM	11
3.5	SEQUENCE DIAGRAM	12
3.6	DATA FLOW DIAGRAM (LEVEL 1)	12

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Social-media platforms have become the primary medium through which people share photographs of their daily lives. The resolution of consumer smartphone cameras has increased to the point where fine biometric detail — fingerprint ridges on a hand gesture, or iris texture in a close-up portrait — is routinely captured without the photographer's awareness. It has been reported that thousands of publicly shared images on platforms such as Instagram contain visible fingerprint detail, and that fingerprints have been successfully lifted from photographs taken from several metres away [1], [2]. A closely related risk exists for the iris: portrait photographs captured on modern smartphones frequently resolve the iris to more than 300 pixels in diameter, which exceeds the resolution required for automated iris recognition, so that iris patterns visible in ordinary portraits can, in principle, be used for biometric matching [3].

Unlike a password, a fingerprint or an iris pattern cannot be revoked once it has been leaked; the same trait is used for the rest of a person's life across banking, government, and device-authentication systems [1], [2]. This irreversibility is what distinguishes biometric privacy from ordinary data privacy and motivates a dedicated, preventive approach rather than a purely reactive one.

1.2 PROBLEM STATEMENT

Given an arbitrary photograph that a user is about to publish on a social-media platform, there is currently no readily available, automated mechanism to inform the user that the image may expose a usable fingerprint or iris pattern. The problem addressed in this project is: given an RGB photograph supplied by a user at upload time, determine (i) whether a hand/finger region and/or an eye/iris region is present, (ii) whether the visible region is of sufficient quality and size to constitute a plausible biometric exposure, and (iii) a risk score and category (Low/Medium/High) that communicates this exposure to the user before the image is posted.

It is important to state explicitly what this project does not attempt. The system is not required to identify the person in the photograph, to authenticate a fingerprint or iris against an enrolled template, to match one biometric sample against another, or to reconstruct a usable biometric template from the image. The scope is strictly limited to exposure detection and privacy-risk estimation.

1.3 OBJECTIVES

- To study and analyse how fingerprint and iris biometric information can be unintentionally exposed in ordinary, casually captured photographs.
- To design a computer-vision pipeline that localises candidate fingerprint (hand/fingertip) and iris (eye) regions in an uploaded image using landmark-detection models.
- To evaluate the quality of each localised region (sharpness, contrast, resolution, size) in order to judge whether it constitutes a genuine biometric exposure rather than a mere detection.
- To propose a Vision Transformer (ViT)-based exposure classifier that analyses a biometric region-of-interest and outputs an exposure-level estimate.
- To design a risk-scoring engine that produces an independent fingerprint risk score, an iris risk score, and an overall privacy-risk score, each mapped to a Low/Medium/High classification.
- To integrate the above into a pre-upload scanning feature inside a social-media-style photo-sharing interface, so that the risk assessment is shown to the user before the image is posted.
- To clearly separate exposure assessment (the scope of this project) from actual biometric-region protection (blurring, pixelation, or masking), which is identified as future work.

1.4 SCOPE

The scope of the present work (Module 1 of the BioShield system) covers pre-upload biometric exposure assessment only. It includes: image upload and preprocessing; hand/fingertip detection and iris/eye detection using landmark-based computer-vision models; image-quality analysis of the detected regions; a proposed Vision-Transformer-based exposure classifier; a risk-scoring engine and Low/Medium/High classification; and a demonstration social-media-style interface that shows the scan result to the user prior to posting.

The scope explicitly excludes biometric identification, authentication, or matching of any kind; reconstruction of a fingerprint or iris template from an image; and the actual protective transformation of a flagged region (blurring, pixelation, or masking), which is planned as a separate Module 2 outside the current phase of the project.

1.5 SECURITY REQUIREMENTS / SECURITY CONSIDERATIONS

- Data minimisation: the system should process an uploaded image only for the duration of the scan and should not retain a biometric template extracted from it.
- No identity linkage: a scan result must not be linked to, or used to build, an identity-linked biometric database of the user or of any other person appearing in the photograph.
- Confidentiality of scan results: risk scores and detected-region data stored for demonstration purposes (e.g., in the prototype's Supabase/SQLite store) must be access-controlled and should not be exposed through an unauthenticated API endpoint.
- Input validation: uploaded files must be validated for type, size, and integrity before being passed to the detection and inference pipeline, to prevent malformed or malicious files from being processed.
- Principle of least privilege: the backend analysis service should only be granted the permissions it needs to read the uploaded image and write the scan result, and should not require broader access to a user's account or device.
- Transparent limitations: because the system assesses risk rather than guaranteeing protection, the interface must communicate clearly that a Low-risk result is not an absolute guarantee of safety, consistent with the system's defensive, decision-support framing.

1.6 SDG ALIGNMENT

The project aligns with the United Nations Sustainable Development Goals (SDGs) in the following ways. Table 1.1 summarises the relevant goals, targets, and the corresponding contribution of this project.

Table 1.1: SDG Alignment of the Proposed Project

S. No.	SDG No. & Goal	Relevant SDG Target	Project Contribution / Alignment
1	SDG 9: Industry, Innovation and Infrastructure	Target 9.5 – Enhance scientific research and upgrade technological capabilities	Applies Vision Transformer-based computer vision to a novel, under-addressed stage of biometric privacy protection.
2	SDG 16: Peace, Justice and Strong Institutions	Target 16.10 – Ensure public access to information and protect fundamental freedoms	Helps protect individuals' biometric privacy, a fundamental aspect of personal data

S. No.	SDG No. & Goal	Relevant SDG Target	Project Contribution / Alignment
			protection, before information is shared publicly.
3	SDG 4: Quality Education	Target 4.7 – Promote knowledge and awareness relevant to sustainable development and global citizenship	Raises user awareness of biometric privacy risk through an interpretable, explainable risk assessment at the point of sharing.

CHAPTER 2

LITERATURE REVIEW

2.1 EXISTING SYSTEM WITH SECURITY APPROACHES

Malhotra et al. [1] were among the first to study "finger-selfies" and proposed a white-box anonymisation method for casually captured fingertip photographs, but noted that protection under an unknown (black-box) recognition model remained an open problem. Li et al. [2] addressed this gap with FingerSafe, a hierarchical perceptual noise-injection framework that perturbs the fingerprint orientation field for black-box transferability while suppressing low-level local-contrast changes for visual naturalness; FingerSafe reports up to 94.12% protection in digital settings and up to 68.75% protection when images are round-tripped through Twitter and Facebook.

Barni et al. [3] demonstrated that iris recognition can succeed on images collected from public websites even when face recognition on the same images fails, and proposed a GAN-based de-identification pipeline that replaces the iris texture using the rubber-sheet model while preserving eye colour and reflection cues for visual realism, reducing the achievable equal-error rate to close to 50% (i.e., near random). Fawkes [9], LowKey [10], TIP-IM [11], and Unlearnable Examples [8] extend the adversarial-protection idea to face images, aiming respectively to poison unauthorised facial-recognition training, resist commercial face-recognition APIs, generate encrypted identity masks, and make training data unusable for a downstream model.

A separate line of research addresses biometric template protection rather than raw images: cancelable biometrics [18], [19] and biometric cryptosystems transform an enrolled template so that the original biometric feature cannot be recovered even if the protected template is stolen, and surveys [12], [13], [14] summarise the trade-offs between recognition accuracy and irreversibility in such schemes. Outside the biometric-specific literature, systems such as PicAlert [20] have studied automated classification of whether a shared photograph is generally "private" or "public", establishing that automated, pre-sharing privacy classifiers are feasible and useful.

2.2 RESEARCH GAPS

Existing fingerprint- and iris-privacy research clusters around two stages of the biometric lifecycle: (a) protecting an image that has already been captured, using adversarial perturbation

[2], [8]–[11] or de-identification [3], [17], and (b) protecting a template that has already been extracted, using cancelable-biometric or cryptosystem techniques [12], [13], [18], [19]. Neither stage addresses the moment at which an ordinary user is about to share a photograph that was never intended to contain biometric information. General image-privacy tools such as PicAlert [20] operate at the sharing stage but do not reason about fingerprint or iris regions specifically.

Table 2.1: Positioning of the Proposed System Relative to Existing Approaches

Approach	Focus	Gap Relative to Proposed System
Adversarial perturbation [2], [8]–[11]	Protect an already-captured image from recognition models	Acts after capture; does not warn the user before sharing
Biometric de-identification [3], [17]	Replace biometric region with synthetic content	Acts after capture; modifies the image outright
Cancelable biometrics / template protection [12], [13], [18], [19]	Protect an enrolled template from theft or reversal	Assumes a biometric enrolment pipeline already exists
General image privacy classifiers [20]	Flag broadly "private" images before sharing	Not biometric-region aware; no fingerprint/iris localisation
Proposedsystem (BioShield)	Pre-upload fingerprint and iris exposure risk assessment	Assessment only in current scope; protective transformation is future work

The research gap addressed by this project is therefore the absence of a pre-upload, biometric-region-aware risk-assessment tool that (i) localises candidate fingerprint and iris regions in an arbitrary photograph, (ii) evaluates whether the localised region is of sufficient quality to constitute a real exposure risk, and (iii) reports an interpretable, per-region and overall risk score to the user at the point of sharing, before any protective transformation is applied.

2.3 PROPOSED SYSTEM

2.3.1 Overview of the Proposed System

The proposed system, BioShield, is a pre-upload biometric privacy assessment framework embedded in a social-media-style photo-sharing workflow. It combines MediaPipe-based landmark detection for hand/fingertip and face/eye/iris localisation, OpenCV-based image-quality analysis, and a Vision-Transformer-based exposure classifier into a risk-scoring engine that produces a fingerprint risk score, an iris risk score, and an overall privacy risk score, each on a 0–100 scale and mapped to a Low/Medium/High category.

2.3.2 Objectives of the Proposed System

- Detect candidate fingerprint and iris regions in an uploaded photograph without performing biometric identification.
- Distinguish detection confidence (a region is present) from privacy risk (the region is of a quality that constitutes exposure).
- Provide an interpretable Low/Medium/High risk category and a visual explanation before the image is shared.
- Provide the analysis required for a subsequent protection module (Module 2) without performing the protective transformation itself.

2.3.3 Advantages of Proposed System

- Acts proactively, before an image is shared, rather than reactively after it has already been published.
- Keeps detection confidence and privacy-risk scoring conceptually distinct, improving the interpretability of the result.
- Does not require a biometric enrolment pipeline, a reference database, or identity information about the user.
- Modular design cleanly separates exposure assessment from protection, allowing the protection module (Module 2) to evolve independently.
- Uses lightweight, real-time landmark-detection models (MediaPipe) that are practical for an interactive, client-facing scanning workflow.

CHAPTER 3

SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

At the highest level, BioShield accepts an image from the user, runs it through parallel fingerprint and iris analysis branches, and combines their outputs into a single risk report, as shown in Fig. 3.1.

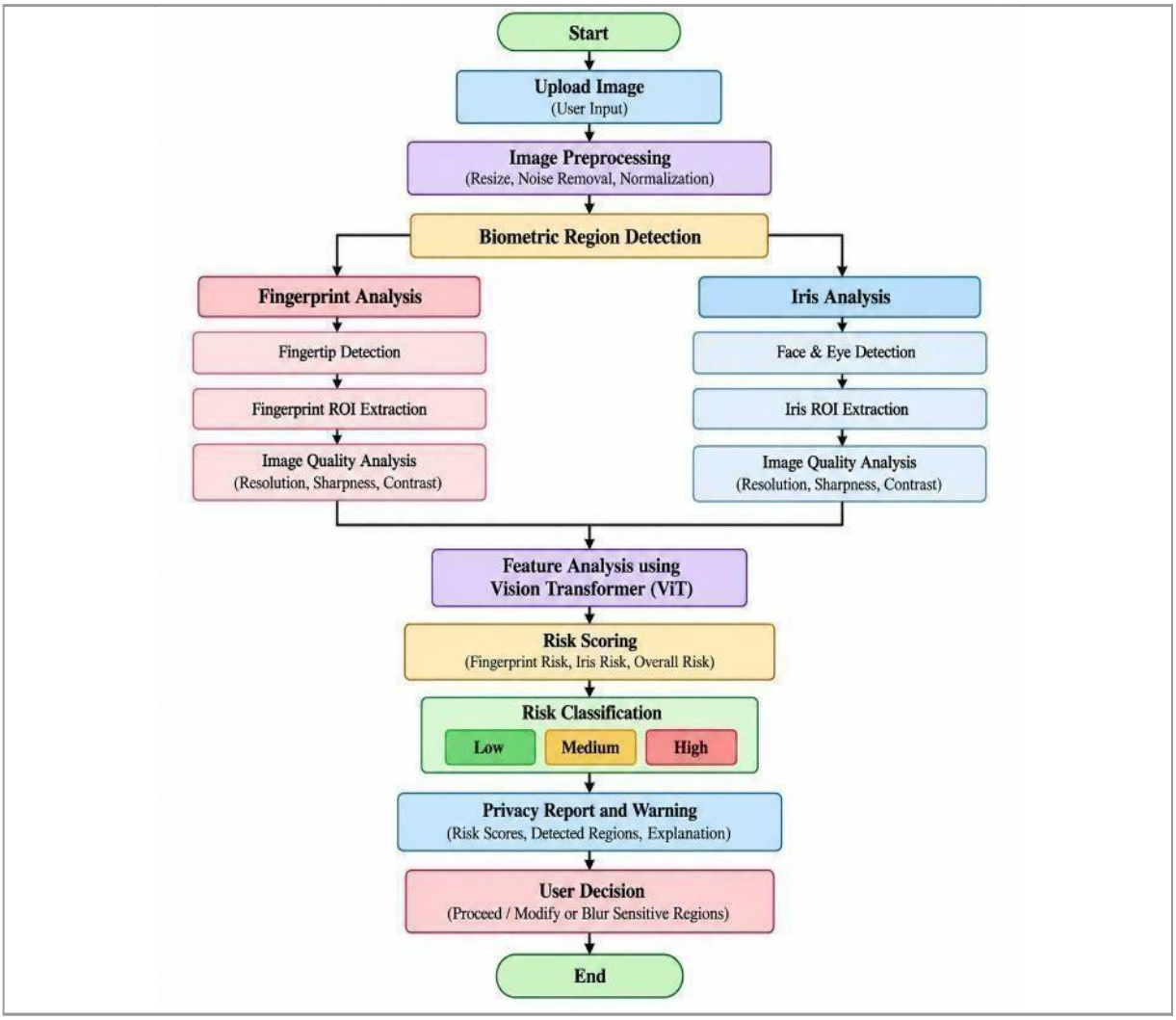


Fig. 3.1: Simplified system architecture of BioShield showing the two parallel analysis branches feeding a common risk-scoring engine.

The detailed architecture (Fig. 3.2) decomposes the system into fifteen functional components: (1) the user interface, (2) the image-upload module, (3) image preprocessing, (4) hand/finger detection, (5) fingerprint ROI extraction, (6) face/eye/iris detection, (7) iris ROI extraction, (8) image-quality analysis, (9) the ViT-based exposure-assessment model, (10) the risk-scoring engine, (11) privacy-risk classification, (12) detection visualisation, (13) risk explanation, (14) scan-result storage, and (15) the Module-2 handoff interface.

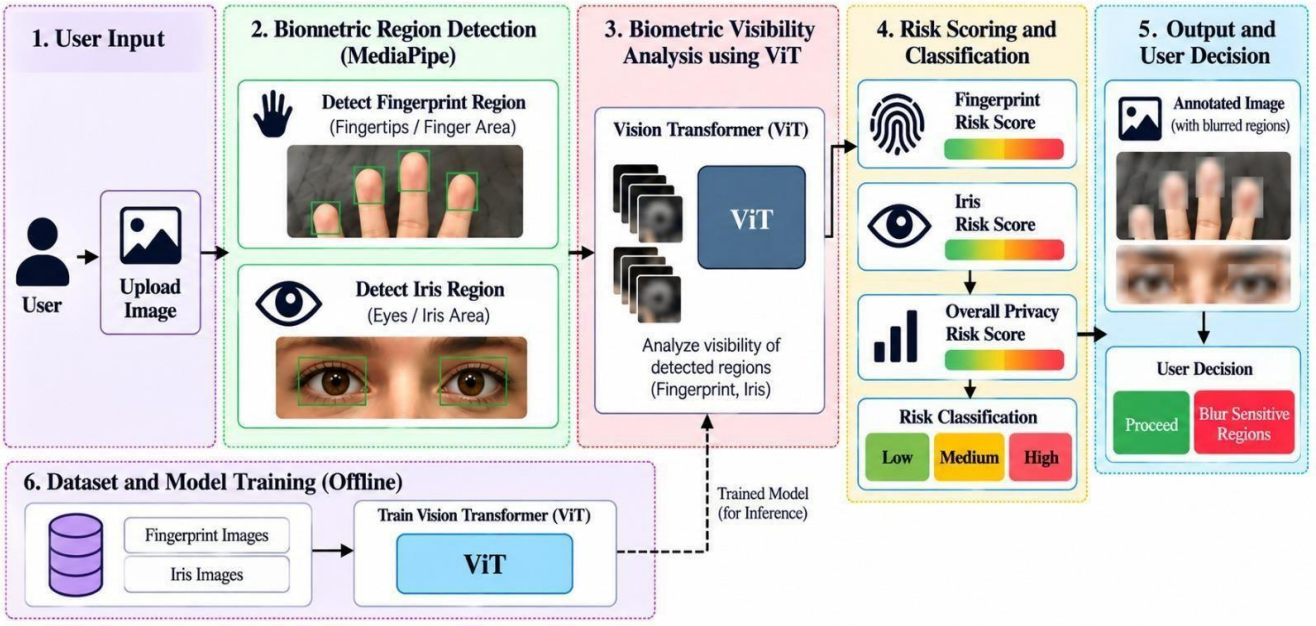


Fig. 3.2: Detailed component-level architecture of BioShield, showing data flow from image upload through both analysis branches to the risk-scoring and visualisation stages.

Table 3.1: Hardware and Software Requirements

Category	Requirement
Processor	Intel Core i5 (10th Gen) or higher / equivalent AMD Ryzen 5 processor
RAM	8 GB minimum (16 GB recommended for model fine-tuning)
GPU	CUDA-enabled GPU recommended for ViT fine-tuning and inference (optional for CPU-only inference)
Camera / Input	Smartphone or webcam-captured RGB images (JPEG/PNG)
Operating System	Windows 10/11, Ubuntu 20.04+, or macOS
Programming Language	Python 3.10+
Libraries / Frameworks	OpenCV, MediaPipe, PyTorch, Flask/FastAPI
Front-end	React / HTML / CSS / JavaScript
Database	Supabase / SQLite
IDE / Tools	VS Code, Jupyter Notebook, Git

3.2 LIST OF MODULES

The system is organised into the following modules, each corresponding to a functional component identified in the architecture diagrams above.

Module 1 – Biometric Detection and Risk Assessment

This module detects potential fingerprint and iris exposure in uploaded images. It analyzes the image,

identifies sensitive biometric regions, calculates a risk score, and classifies the result as **Low, Medium, or High** risk.

Module 2 – Biometric Privacy Protection

This module protects detected biometric regions using **Soft/Micro Local Blur**. The protection is applied only to the sensitive fingerprint and iris regions while maintaining the natural appearance of the image. The user can adjust the protection strength before generating the protected image.

Module 3 – Protection Verification

This module verifies whether the applied privacy protection has sufficiently reduced biometric exposure. The protected image is analyzed again and the original and protected risk levels are compared to determine the effectiveness of the protection.

Module 4 – User, Security and Privacy Management

This module manages user authentication, profiles, scan history, privacy settings, and stored analysis results. It provides users with a privacy dashboard to monitor their biometric protection activities securely.

Module 5 – Advanced AI and Production Enhancement

This module focuses on improving the system using advanced AI-based biometric detection, better risk assessment, evaluation metrics, and administrative monitoring. It is designed to improve the accuracy, scalability, and reliability of BioShield for real-world use.

3.3 DATA FLOW / UML DIAGRAMS

Fig. 3.3 to Fig. 3.6 present the use-case, activity, sequence, and data-flow views of the system.

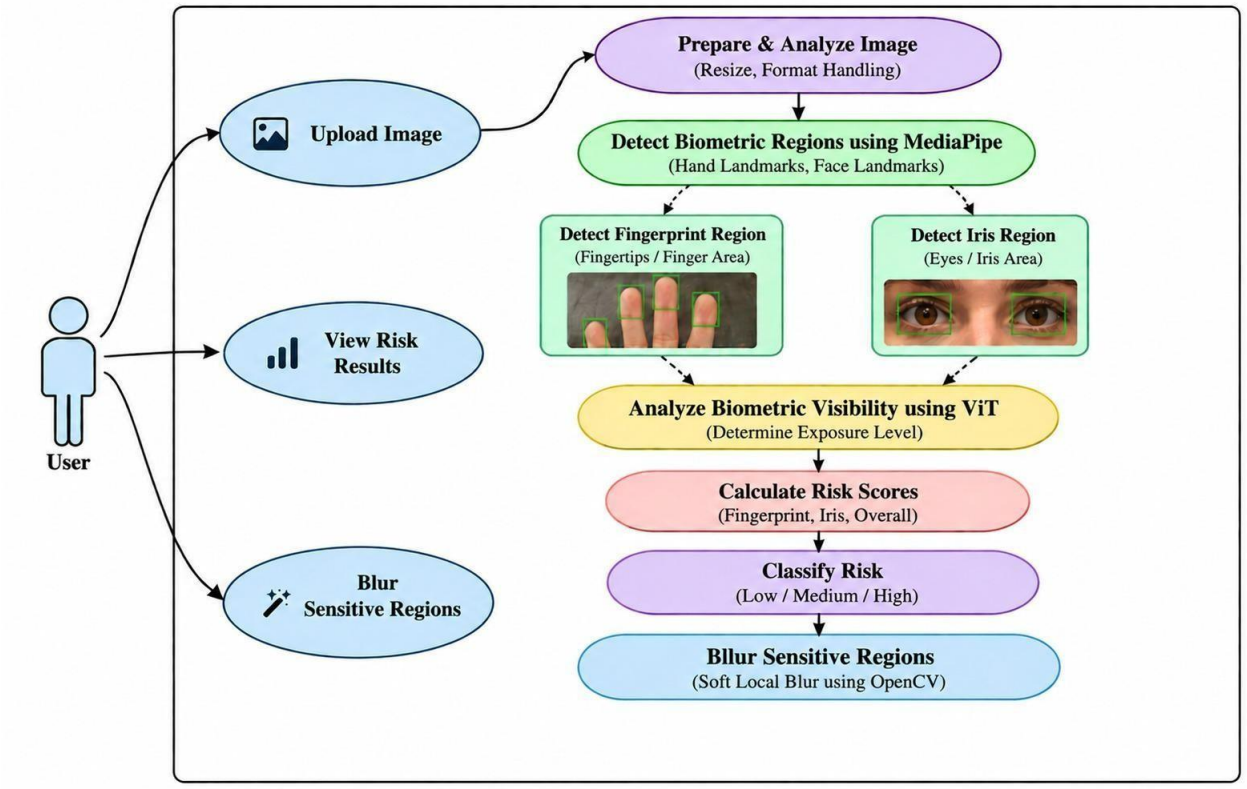


Fig. 3.3: Use Case Diagram – actors (User) and use cases (Upload Image, Scan for Privacy Risk, View Risk Result, Protect Image, Post Anyway, Cancel).

The use-case diagram identifies a single primary actor, the User, who interacts with four principal use cases: “Upload Image”, “Scan for Biometric Privacy”, “View Risk Result”, and “Decide Next Action” (Protect Image / Post Anyway / Cancel). The “Scan for Biometric Privacy” use case internally includes the “Detect Fingerprint Region” and “Detect Iris Region” use cases.

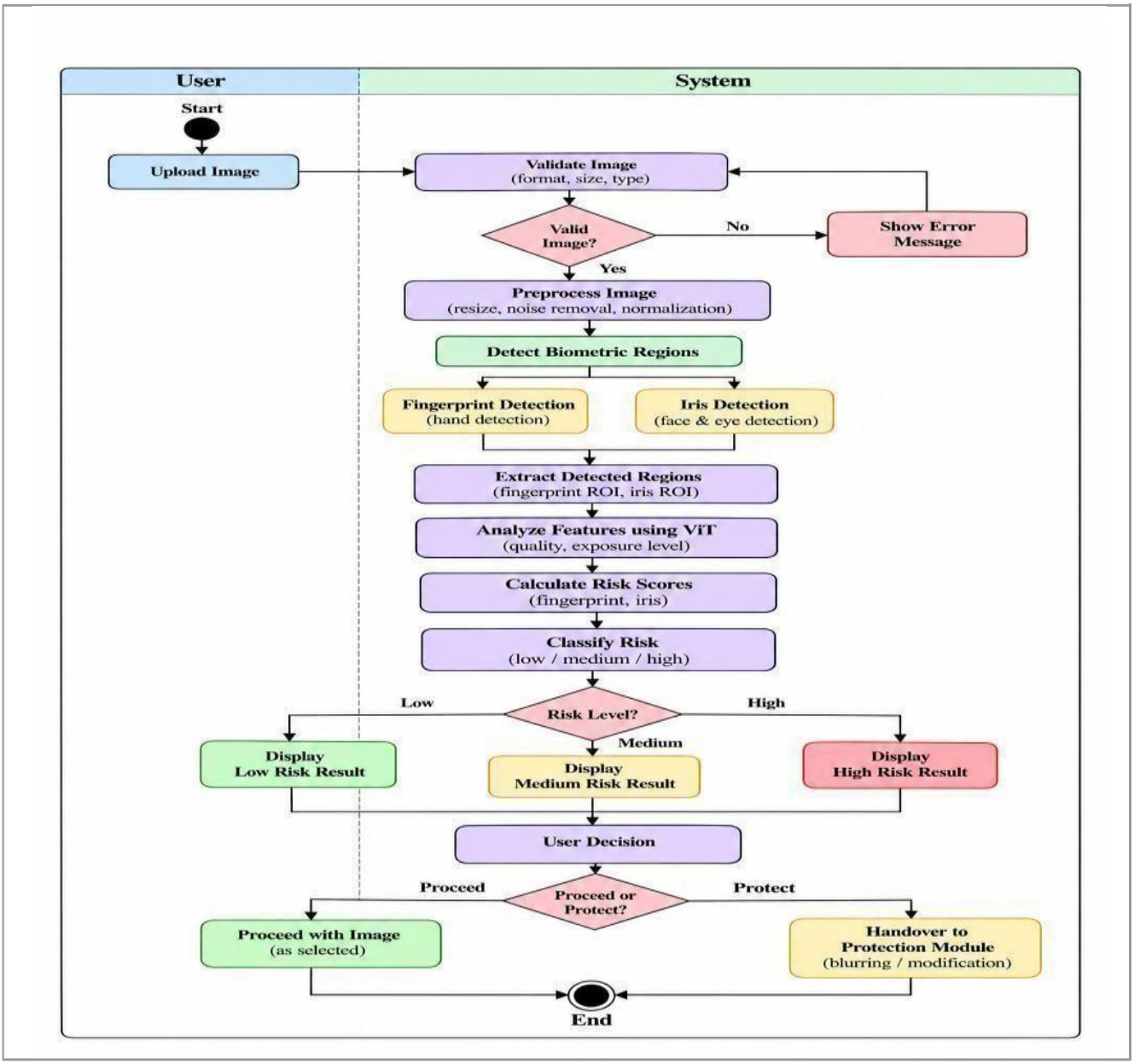


Fig. 3.4: Activity Diagram – end-to-end flow from image upload through detection, quality analysis, ViT classification, risk scoring, and result display.

The activity diagram traces the control flow described in Section 4.2.1: upload → validate → preprocess → parallel hand/face detection → ROI quality check → ViT classification (for qualifying ROIs) → risk score fusion → classification → result display → user decision (protect / post / cancel).

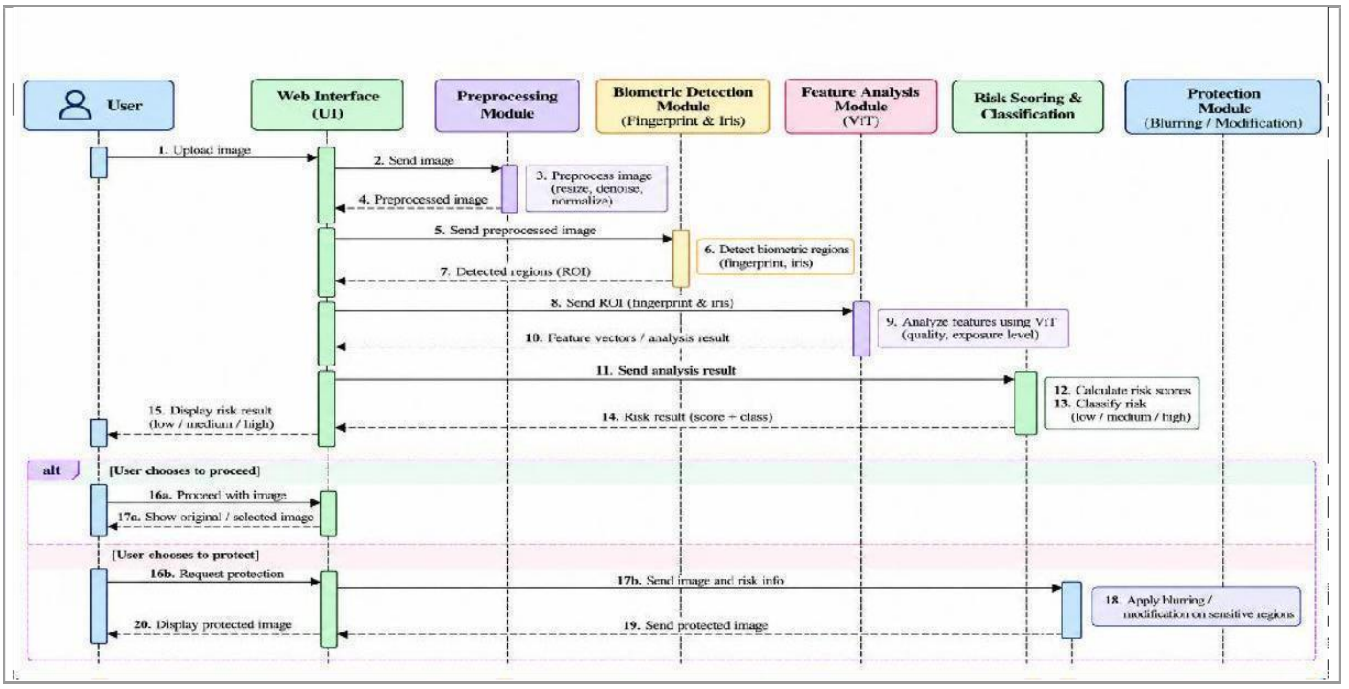


Fig. 3.5: Sequence Diagram – interaction between the User, the Upload Module, the Detection Services, the ViT Inference Service, and the Risk Scoring Engine.

The sequence diagram shows the temporal ordering of calls: the User triggers an upload; the Upload Module forwards the normalised image to the Hand-Detection and Face-Detection services in parallel; each service returns landmark data to the Risk Scoring Engine, which invokes the ViT Inference Service for qualifying ROIs before returning the combined risk report to the User Interface.

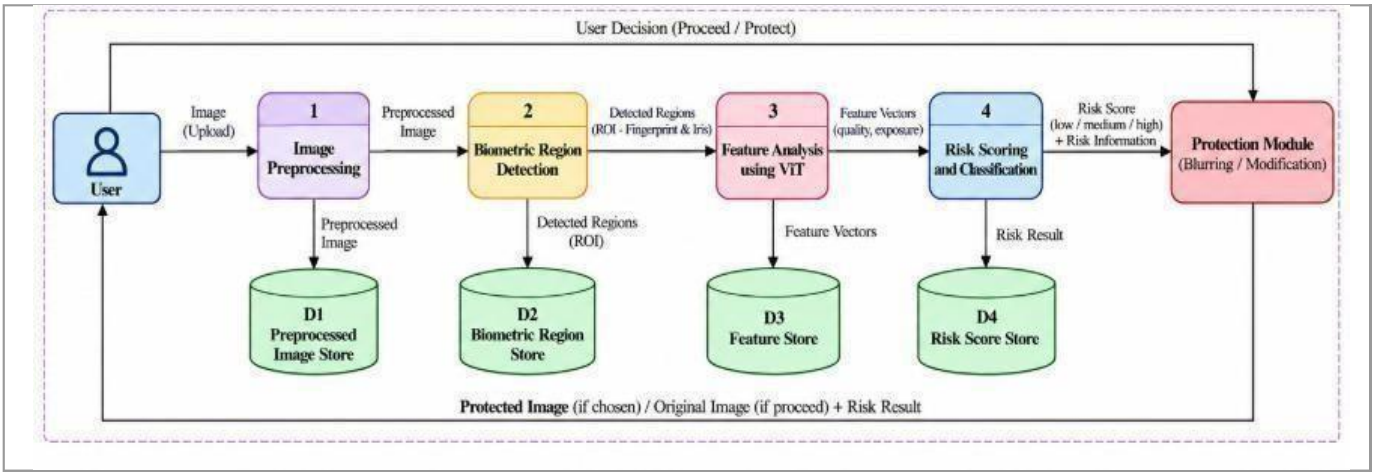


Fig. 3.6: Data Flow Diagram (Level 1) – movement of image data, ROI crops, quality metrics, and risk scores between system components.

3.4 THREAT MODEL AND ATTACK SURFACE

Although BioShield is a defensive privacy tool rather than an authentication system, it still has an attack surface that must be considered. The assets to protect are: (a) the image being scanned, while it is in transit and in memory during processing; (b) the scan result (risk scores and detected-region metadata) stored for the session or demo; and (c) the availability and integrity of the detection and inference pipeline itself.

- Malicious or malformed image uploads: a crafted image file could attempt to exploit a vulnerability in an image-decoding library or in MediaPipe/OpenCV; mitigated by strict input validation and the use of well-maintained, patched libraries.
- Unauthorised access to stored scan results: if the scan-result store (Supabase/SQLite) is not properly access-controlled, another party could read another user's scan history; mitigated by authentication and access control on the storage layer.
- Model evasion: an adversarially crafted image could, in principle, be designed to make the ViT classifier under-report exposure; this is a known limitation of learned classifiers in general and is noted in the limitations of the methodology.
- Denial of service: repeated large uploads could exhaust backend compute resources; mitigated by file-size limits and rate limiting on the upload endpoint.
- Privacy risk of the tool itself: because the tool processes images that may contain real biometric detail, the backend must avoid retaining raw biometric crops beyond the scan session, to avoid becoming, itself, a biometric data store.

3.5 SECURITY CONTROLS

- Data minimisation and ephemeral processing: cropped ROIs are processed in memory for the duration of the scan and are not persisted beyond what the scan-result record requires.
- Input validation: file-type, file-size, and image-integrity checks are performed before any detection or inference step.
- Least-privilege backend design: the analysis service is scoped to read the uploaded image and write the scan result only; it does not require broader account access.
- Access-controlled storage: the scan-result store requires authenticated access, consistent with standard secure API design practice.
- Separation of concerns: the explicit Module 1/Module 2 boundary ensures that any future protective transformation is implemented and reviewed independently of the risk-assessment logic described here.
- Transparent, bounded claims: the system communicates a risk estimate, not a security guarantee, which is itself a control against users over-trusting a Low-risk result.

CHAPTER 4

SYSTEM DESCRIPTION

4.1 DATA COLLECTION

Dataset selection is to be finalised. The proposed strategy uses two categories of imagery: (i) hand/fingertip images at varying distance, angle, and resolution, drawn from publicly available finger-selfie or hand-gesture datasets referenced in the fingerprint-privacy literature [1], [2]; and (ii) facial/eye images at varying resolution, drawn from publicly available face/iris datasets such as those referenced by Barni et al. [3], [16].

Each image is proposed to be labelled with a biometric exposure level (Low/Medium/High) derived from a combination of ground-truth region size, sharpness, and, where available, whether a reference biometric matcher can successfully extract a usable template from the region. Negative samples — images containing no visible hand or eye region, or containing heavily occluded or blurred regions — are required so that the model learns to avoid over-flagging. All data collection is proposed to rely on publicly available, appropriately licensed research datasets and/or consenting volunteers, in line with the ethical practice followed in the reference studies [1], [3], [16]; no biometric data is proposed to be collected beyond what is strictly required for exposure-level labelling, and no attempt is made to build an identity-linked biometric database.

4.2 DATA PREPROCESSING

Uploaded images are first validated for format and minimum resolution, then decoded and normalised for downstream processing. Preprocessing steps performed with OpenCV include colour-space conversion (BGR to RGB), optional resizing to a bounded maximum dimension for consistent processing time, and orientation correction using embedded EXIF metadata where available. A conventional 70/15/15 train/validation/test split is proposed for the labelled ROI dataset, with standard augmentation (rotation, brightness/contrast jitter, and compression-quality variation) applied to the training partition to approximate the range of conditions seen on real social-media uploads.

4.2.1 Algorithms / Methodology

The following algorithms summarise the core processing logic of the proposed system.

```
Algorithm 1: Overall Biometric Privacy Analysis
  Input: image I
  Output: fingerprintRisk, irisRisk, overallRisk, category
```

```
  1. I' <- Preprocess(I)
  2. Hf <- DetectHands(I')
  3. Fe <- DetectFaceEyes(I')
  4. if Hf not empty: fingerprintRisk <- AssessFingerprint(I', Hf)
     else fingerprintRisk <- 0
  5. if Fe not empty: irisRisk <- AssessIris(I', Fe)
     else irisRisk <- 0
  6. overallRisk <- Combine(fingerprintRisk, irisRisk)
  7. category <- Classify(overallRisk)
  8. return fingerprintRisk, irisRisk, overallRisk, category
```

```
Algorithm 2: Fingerprint Exposure Assessment
  Input: image I', hand landmarks Hf
  Output: fingerprintRisk
  1. score <- 0
  2. for each fingertip f in Hf:
         roi <- CropROI(I', f)
         q <- QualityMetrics(roi)
         if q passes threshold:
             p <- ViT_Fingerprint(roi)
             s <- FuseScore(f.confidence, q, p)
         else s <- PartialScore(f.confidence, q)
         score <- max(score, s)
  3. return score
```

```
Algorithm 3: Iris Exposure Assessment
  Input: image I', eye landmarks Fe
  Output: irisRisk
  1. score <- 0
  2. for each eye e in Fe:
         roi <- CropIrisROI(I', e)
         q <- QualityMetrics(roi)
         if q passes threshold:
             p <- ViT_Iris(roi)
             s <- FuseScore(e.confidence, q, p)
         else s <- PartialScore(e.confidence, q)
         score <- max(score, s)
  3. return score
```

```
Algorithm 4: Privacy Risk Calculation
  Input: fingerprintRisk, irisRisk
  Output: overallRisk, category
  1. overallRisk <- w1*fingerprintRisk + w2*irisRisk
  2. if overallRisk < 40: category <- LOW
  3. else if overallRisk < 70: category <- MEDIUM
  4. else: category <- HIGH
  5. return overallRisk, category
```

```
Algorithm 5: ViT-Based Exposure Classification
  Input: ROI crop r (fingerprint or iris)
  Output: exposure probability distribution p
  1. r' <- Resize(r, 224x224)
  2. patches <- SplitPatches(r', 16x16)
  3. z <- LinearEmbed(patches) + PositionalEmbed
```

```
4. z <- Prepend(CLS_token, z)
5. z <- TransformerEncoder(z)
6. p <- Softmax(ClassificationHead(z[CLS]))
7. return p
```

Table 4.1: Privacy Risk Classification

Category	Score Range	Interpretation
Low	0 – 39	Minimal likelihood of exploitable biometric exposure
Medium	40 – 69	Possible exposure; user is advised to review the region
High	70 – 100	Strong likelihood of exploitable exposure; protection recommended

4.2.2 Results and Discussion

Experimental results are to be added after implementation and evaluation are completed. This section is intentionally left as a template so that the following items can be inserted once the corresponding experiments have been carried out, rather than presenting hypothetical numbers as findings.

- Dataset composition and the train/validation/test split actually used.
- Accuracy, precision, recall, and F1-score for the fingerprint-exposure classifier.
- Accuracy, precision, recall, and F1-score for the iris-exposure classifier.
- Confusion matrices for both classifiers.
- Overall Low/Medium/High risk-classification performance, including false-positive and false-negative rates.
- Measured end-to-end processing time per image on the target deployment hardware.
- Comparison against a simple rule-based baseline that uses only detection confidence and region size, without the ViT classifier.

To populate this section, the following experiments need to be performed: (1) finalise and label the dataset described in Section 4.1; (2) fine-tune the fingerprint and iris ViT classifiers; (3) run the full pipeline on a held-out test set and compute the metrics listed above; (4) implement the rule-based baseline for comparison; and (5) measure processing latency on the intended deployment hardware.

CHAPTER 5

CONCLUSION

This project presented BioShield, a proposed AI-based fingerprint and iris privacy protection system that targets a stage of the biometric-privacy pipeline — pre-upload exposure assessment — that is not addressed by existing fingerprint-privacy [1], [2], iris-privacy [3], de-identification [17], or biometric template-protection [12], [13], [18], [19] research. The current implementation (Module 1) combines MediaPipe-based region localisation [5]–[7], OpenCV-based image-quality analysis, and a proposed Vision-Transformer-based [4] exposure classifier into a risk-scoring engine that reports fingerprint, iris, and overall privacy risk on a 0–100 scale with an interpretable Low/Medium/High classification, inside a social-media-style upload workflow.

The system explicitly does not perform biometric identification, authentication, matching, or reconstruction. The threat model, security controls, architecture, and methodology described in this report are intended to guide the completion of the ViT training, dataset labelling, and quantitative evaluation described in Chapter 4, after which the Results and Discussion section can be populated with actual experimental findings. Future work includes implementing Module 2 (selective blurring, pixelation, or masking of flagged regions), empirically validating the proposed risk-scoring weights, and extending the evaluation to on-device inference latency for a fully client-side pre-upload scan.

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APPENDIX

This appendix is reserved for supporting material that will be added as the implementation progresses: representative source-code listings for the preprocessing, detection, and risk-scoring modules; sample configuration files; and additional screenshots of the social-media-style upload and scan-result interface beyond those referenced as Fig. 3.1–Fig. 4.4 in the main report.

- **Sample source code**

Analysis and Risk Calculation

```
def calculate_risk(fingerprint_score, iris_score):
overall_score = max(
    fingerprint_score,
    iris_score
)

if overall_score >= 70:
    risk_level = "HIGH"
elif overall_score >= 40:
    risk_level = "MEDIUM"
else:
    risk_level = "LOW"

return {
    "fingerprint_score": fingerprint_score,
    "iris_score": iris_score,
    "overall_score": overall_score,
    "risk_level": risk_level
}
```

Biometric Region Detection

```
def detect_biometric_regions(image):
    regions = []

    hands = detect_hands(image)

    for hand in hands:
        for fingertip in hand.fingertips:
            region = {
                "type": "fingertip",
                "x": fingertip.x,
                "y": fingertip.y,
                "width": fingertip.width,
                "height": fingertip.height
            }

            regions.append(region)

    iris_regions = detect_iris(image)

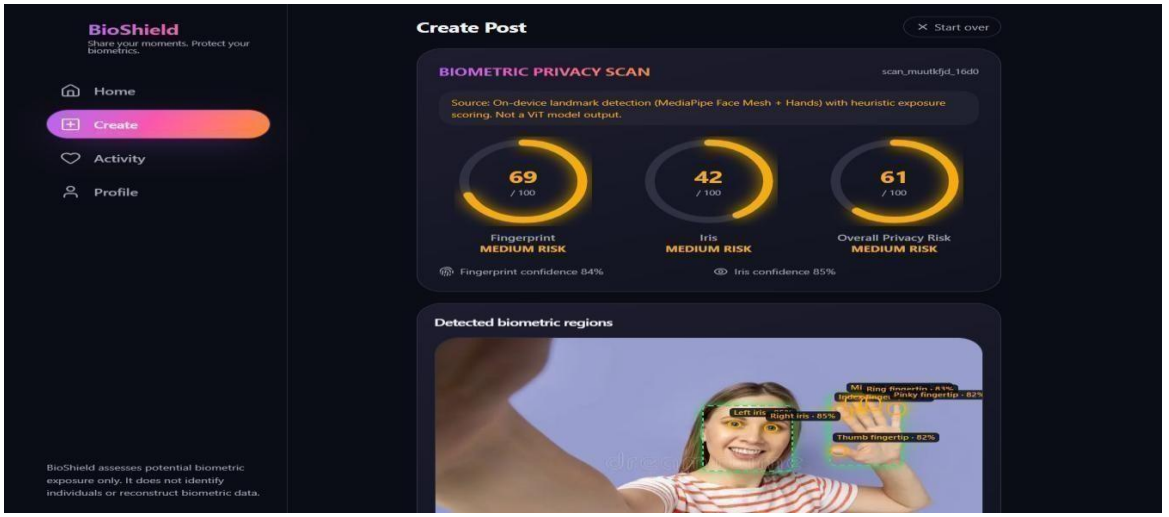
    for iris in iris_regions:
        regions.append({
            "type": "iris",
            "x": iris.x,
            "y": iris.y,
            "width": iris.width,
            "height": iris.height
        })

    return regions
```

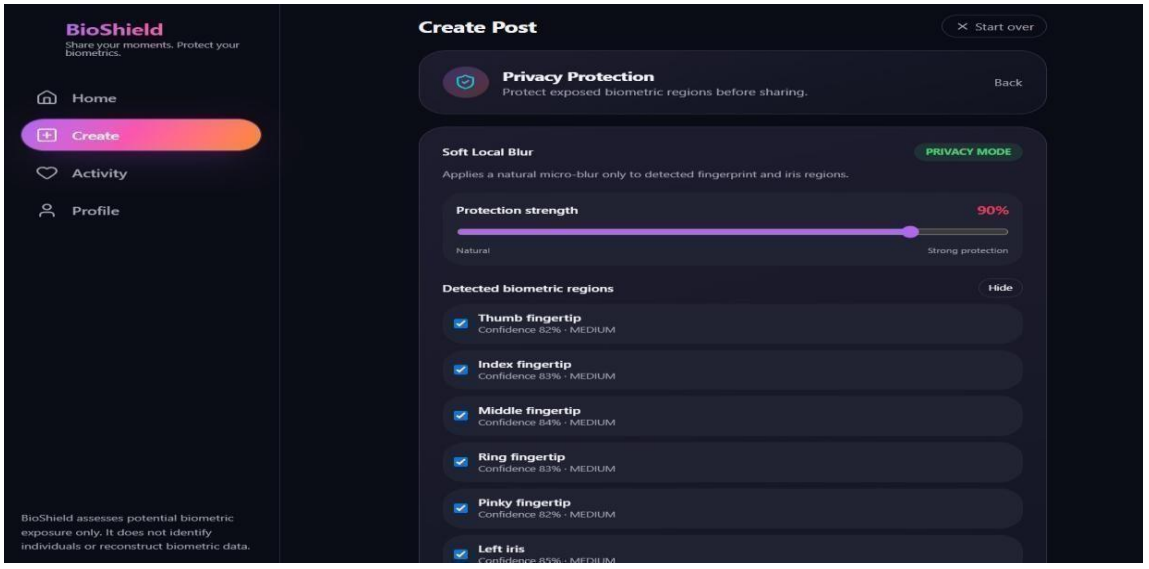
- **Additional interface screenshots**



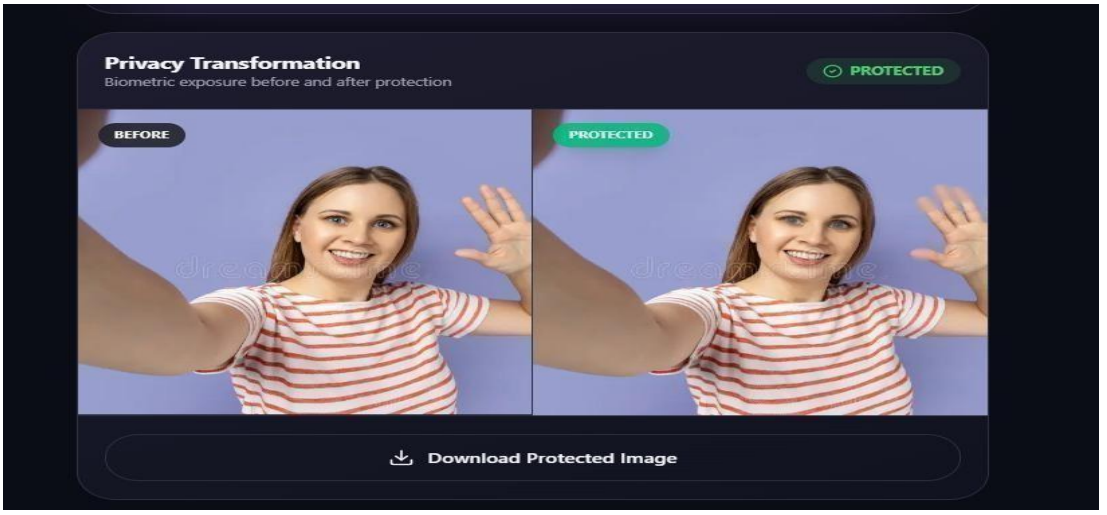
Home Page



Biometric privacy scan



Privacy Protection



Protected image